EE-222 Data Structures and Algorithms

BS(CE)-2k21

LAB REPORT #2

Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

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Lab Instructor.
Kaynat Rana

Experiment # 2

Implementing the Concept of Friend Functions and Friend Classes

Objective:

Objective of this lab session is to reinforce some basic programming concepts regarding to static members, static member functions, friend functions and friend class.

Equipment Used:

Visual studio C++

Procedure:

Friend Functions and Friend Classes:

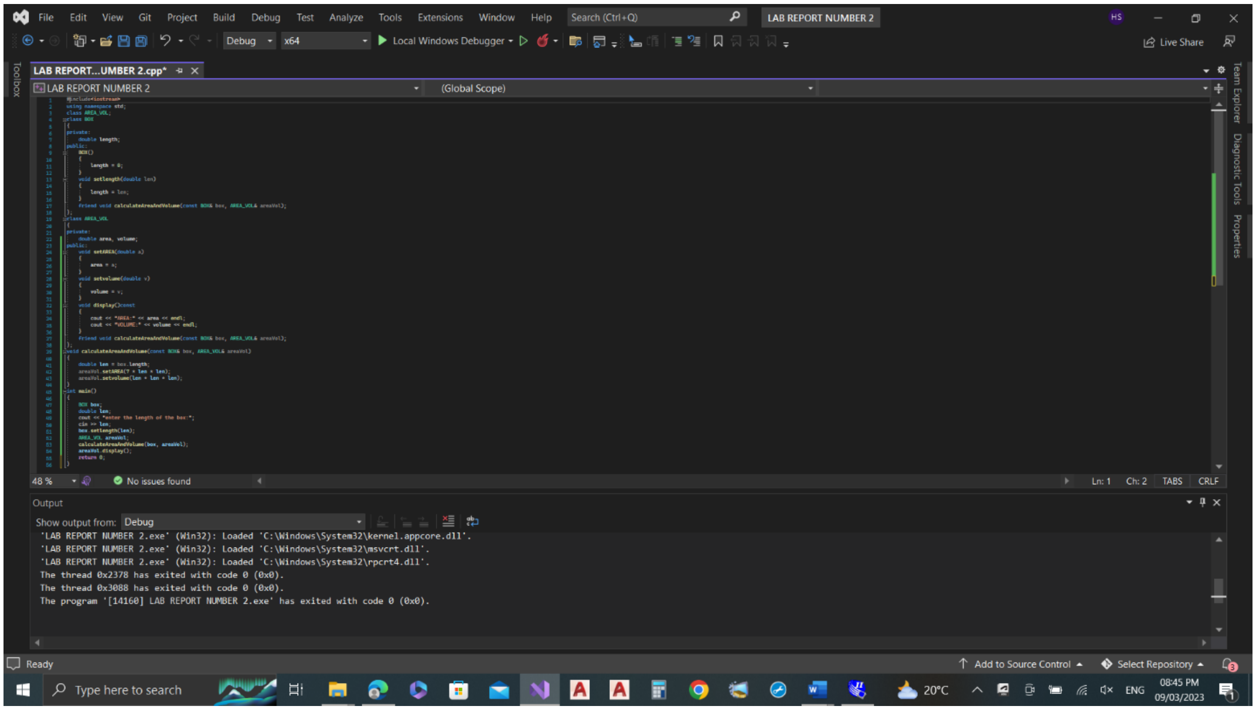
A friend function, that is a " friend" of a given class, is a function that is given the same access as methods to private and protected data. A friend function cannot be member function of the class. Friend Class can access private and protected members of other class in which it is declared as friend. In case of friend class, all of its members are friends. A friend function of a class is defined outside that class; scope but it has the right to access all private and protected members of the class. Even though the prototypes for friend functions appear in the class definition, friends are not member functions.

Tasks

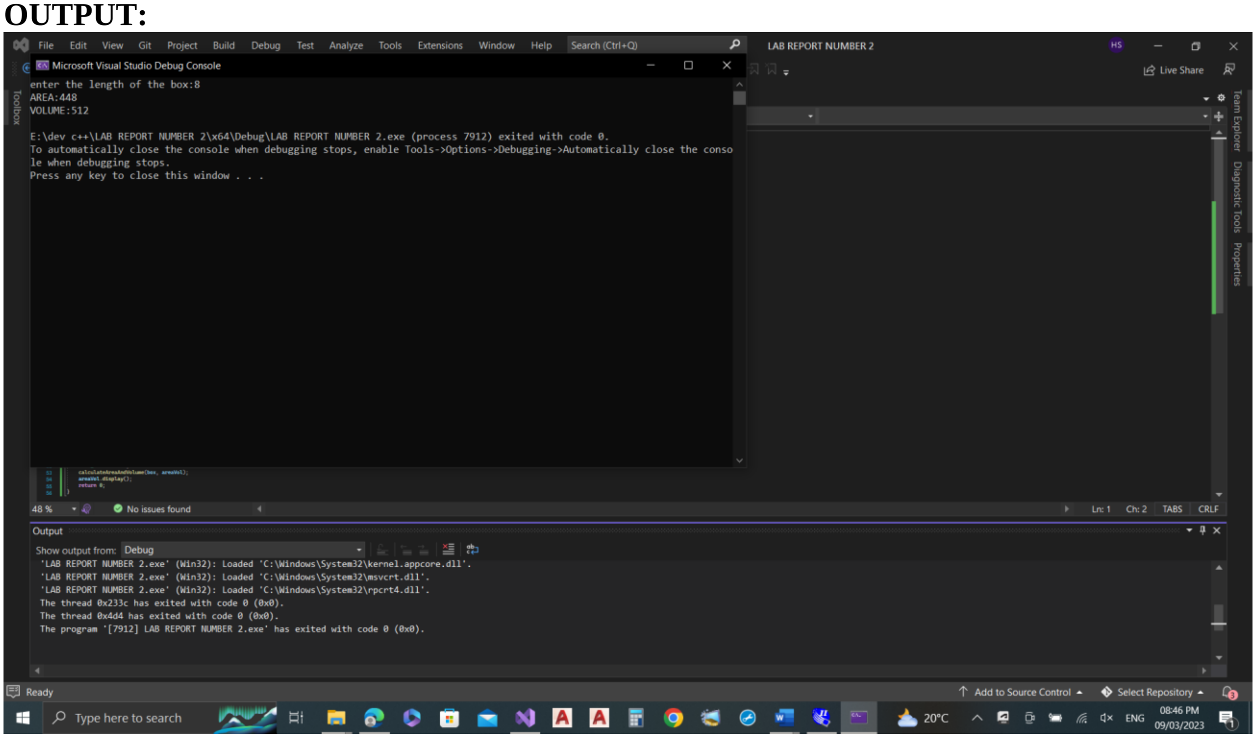
Task 1:

Create and implement a program in C++ to define a BOX class and an AREA_VOL class. In the class BOX, data member should be length. Member function should set the value of data member (user defined or by-passing argument to the member function). Calculate and display the area and volume of the box in AREA_VOL class using friend function. (Note: class AREA_VOL should not be friend class of BOX, but only member function/functions of class AREA_VOL can be friend function/functions of class BOX).

CODE:



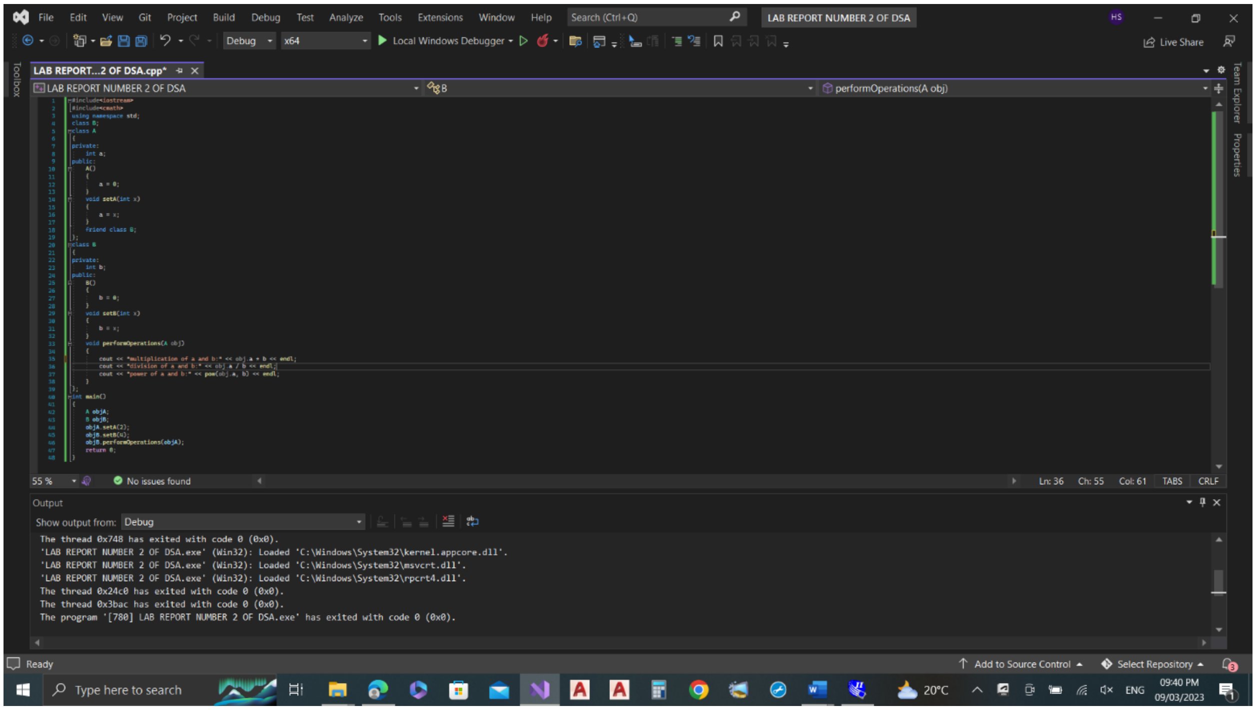
OUTPUT:



TASK 2:

Create and Implement program in C++ to define 2 classes i.e. A and B. Declare an integer ‘a’ variable in class A and an integer ‘b’ in class B. B class should be friend of class A. Division (a/b), power (a^b) and multiplication (a*b) of variables a and b must be performed using member function/ functions of class B.

CODE:



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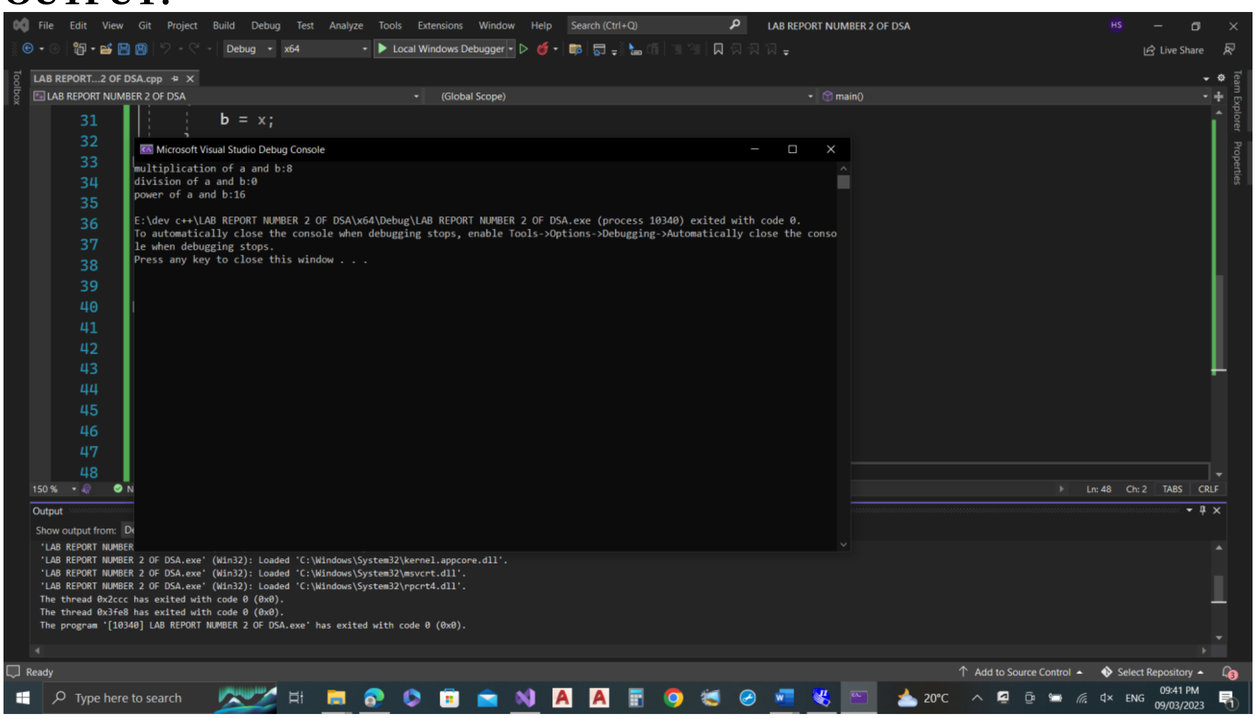
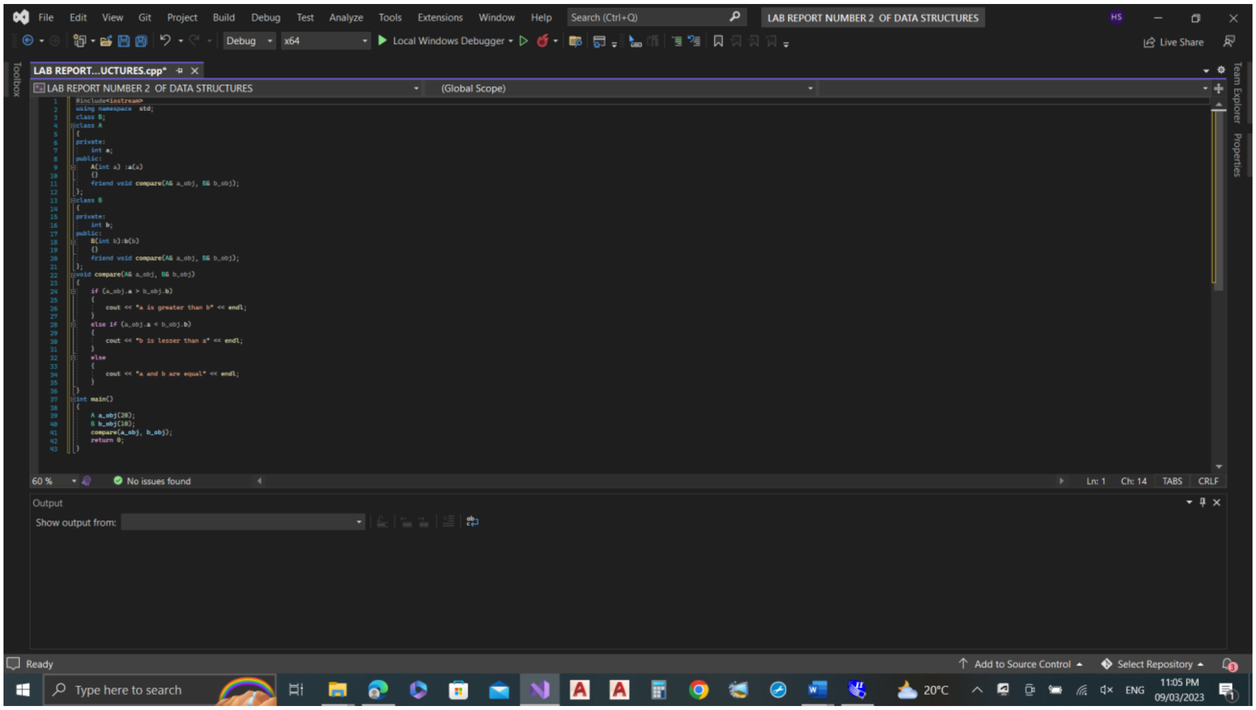


Task 3:

Create and Implement program in C++ to define 2 classes i.e. A and B. Declare an integer ‘a’ variable in class A and an integer ‘b’ in class B. B and A class should not be friend classes. Display the greater number among ‘a’ and ‘b’ using friend function of class A and B. (Note: friend function should not be member function of either class A or B)

CODE:



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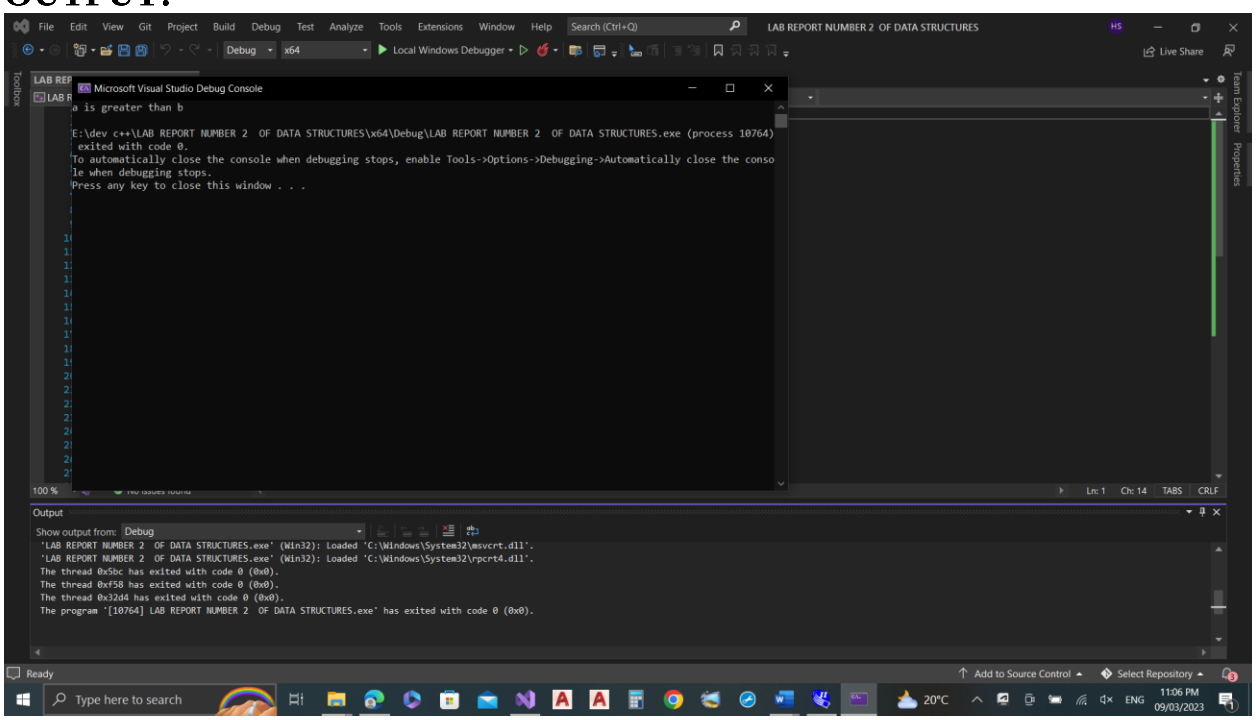

Conclusion:

In conclusion, friend functions and friend classes are powerful tools in object-oriented programming that allow non-member functions and non-member classes to access private and protected members of a class.